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17 OCTOBER 1968
SUPERSEDING
MIL-T-9801B
15 October 1962

MILITARY SPECIFICATION

TRANSMITTER GROUP OA-2857/GSQ-29

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification establishes the production requirements for a transmitter group designated Transmitter Group OA-2857/GSQ-29.

1.2 This group is an expendable airborne signal source that consists of a transmitter and a controller unit, which is used with Scoring System, Vector Miss Distance AN/GSQ-29.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

Federal

FED-STD-595	Colors

FSC 6920

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Military

MIL-STD-130 Identification Marking of US Military Property

DRAWINGS

Air Force

65H32142 Transmitter Group OA-2857/GSQ-29 Scoring System (Matts) Model T9 (6A) (7A) (8A)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Preproduction. This specification mades provision for preproduction testing (see 4.3 and 6.2).

*3.2 Components. The Transmitter Group OA-2857/GSQ-29 shall consist of the following major components:

Item No	Quantity	Description	Part or Dwg No
1	1	Transmitter Unit	24205-1
2	1	Controller Unit	24206-1

3.3 Materials, parts, and processes. Materials, parts, and processes, shall be in accordance with MIL-E-5400, except as otherwise specified herein.

3.3.1 Color. Exterior surfaces shall be painted with at least two coats of lusterless enamel, black number 37038 per FED-STD-595. Surfaces shall be baked at 250° Fahrenheit (F) for one hour between each coat.

*3.4 Design and construction. Detail design and construction of the Transmitter Group OA-2857/GSQ-29 shall be in accordance with Drawing 65H32142 and this specification.

*3.5 Performance. The transmitter group described herein shall have the following performance characteristics:

a. Radiofrequency (RF) power output: Minimum of 7 watts into a 50 ohm load at 28 volt direct current (dc) input.
b. Carrier frequency: As specified in contract (see 6.2):
   (1) Interceptor aircraft: 221 megahertz (MHz) ± 11 kilohertz (KHz).
   (2) Drone target: 224 MHz ± 11 KHz.
   (3) Missile: 225 MHz ± 11 KHz.

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c. Carrier frequency rate of change: Maximum frequency variation of 500 cycles per second (Hertz, Hz).
d. Modulation frequency: 20 KHz plus or minus 250 Hz with a minimum modulation index of 0.7.
e. Input power: 2.4 amperes maximum at 28 volts dc.
f. Input voltage: 28 volts dc.
g. Operational life: 500 hours on external 28 volts supply; 10 minutes on accessory battery. Battery voltage shall not drop below 27 volts dc during this time period.

*3.6 Service requirements, environmental. The performance requirements shall be met when the transmitter group is subjected to the tests described in 4.5.3.2 and 4.5.3.3.

3.7 Weight. Weight of the transmitter group shall not exceed 5 pounds.

3.8 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The nomenclature shall be Transmitter Group OA-2857/GSQ-29.

3.9 Workmanship. The transmitter group shall be manufactured in a thoroughly workmanship manner. The transmitter group shall be free from any defects that might affect appearance, maintainability and functionability.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection of the transmitter group shall be classified as follows:

a. Preproduction tests.
b. Acceptance tests.

4.3 Preproduction testing

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4.3.1 Preproduction test samples. Preproduction test samples shall consist of five of the OA-2857/GSQ-29s representative of the production equipment. Samples shall be identified with the manufacturer's part number and such other information as may be required by the procuring activity.

4.3.2 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.5, Test Methods.

4.4 Acceptance tests. Acceptance tests shall consist of the following:

*4.4.1 Individual tests. Each transmitter group shall be subjected to the following tests described in 4.5, Test Methods.

a. Examination of product.
b. Functional tests.
c. Electromagnetic compatibility tests.

4.5 Test methods

4.5.1 Examination of product. Each item shall be visually examined for workmanship, identification and operational marking, and conformance to the specific design requirements of the drawings (see 3.9).

*4.5.2 Functional tests. Each transmitter group shall be connected to operate under normal room ambient conditions and operated to determine that the transmitter and controller function is in accordance with the requirements of this specification. Special service tools needed to make necessary adjustments during these tests are described in 6.3 of this specification. The following tests shall be made with the equipment connected as shown in figures 1 and 2. Block diagrams of the controller and transmitter units are shown in figures 3 and 4 respectively.

4.5.2.1 RF power output. Set dc source to 28 volts and measure transmitter output power for a period of 1 hour. Power output shall be not less than 7 watts during this entire period.

4.5.2.2 Carrier frequency error. This test shall be run concurrently with test 4.5.2.1 above. Maximum error permitted shall be 11 KHz.

*4.5.2.3 Carrier frequency rate of change. This test shall be run concurrently with test 4.5.2.1. The maximum output frequency rate of change shall not be more than 500 Hz. After transmitter warm-up (approximately ten minutes of operation) the frequency change will decrease to less than 150 Hz.

*4.5.2.4 Modulation frequency. Measure and record modulation frequency with input voltage at 28 volts dc. Maximum modulation frequency error permitted shall be 250 Hz.

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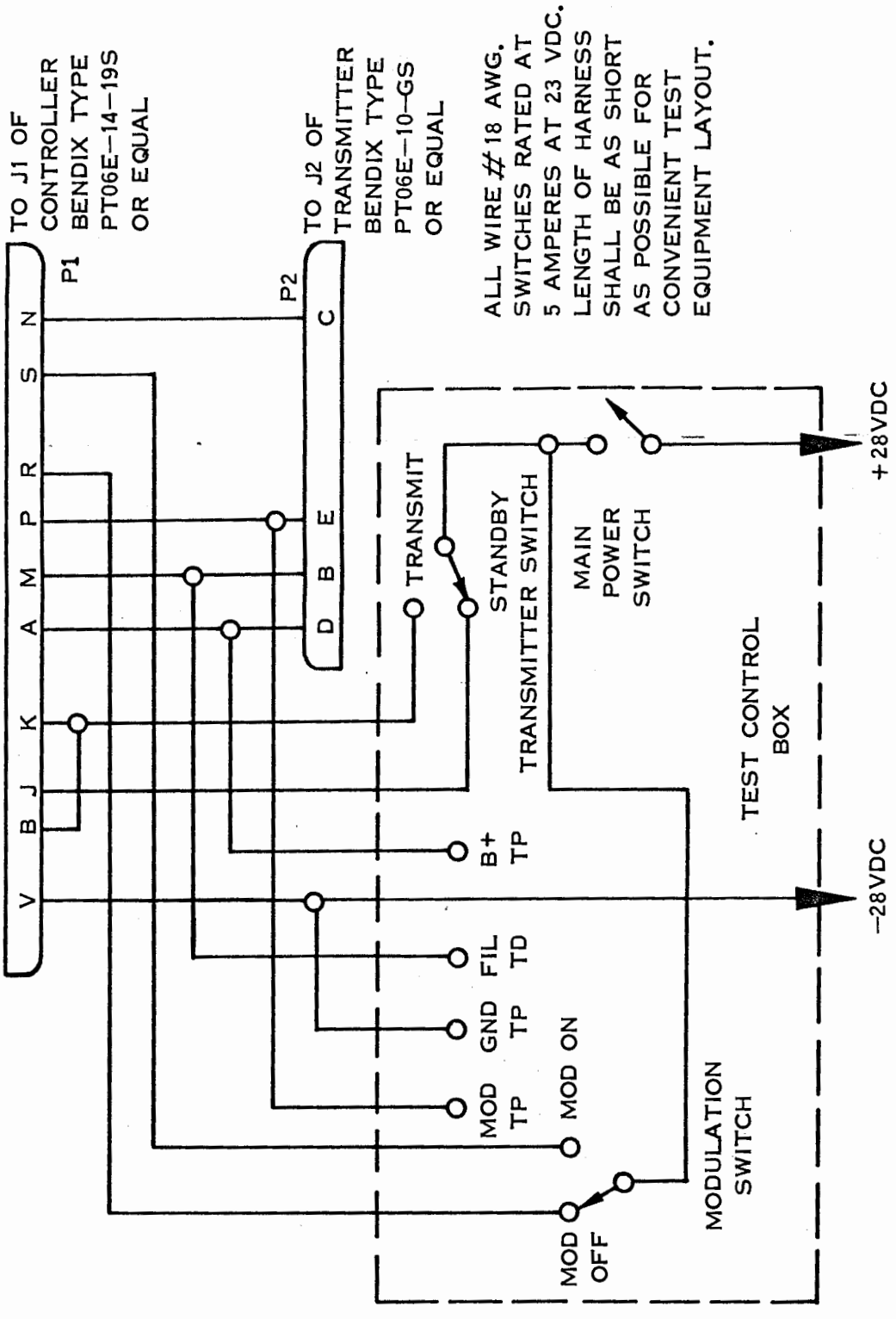


Figure 1. Test Control Box and Test Harness Wiring Diagram

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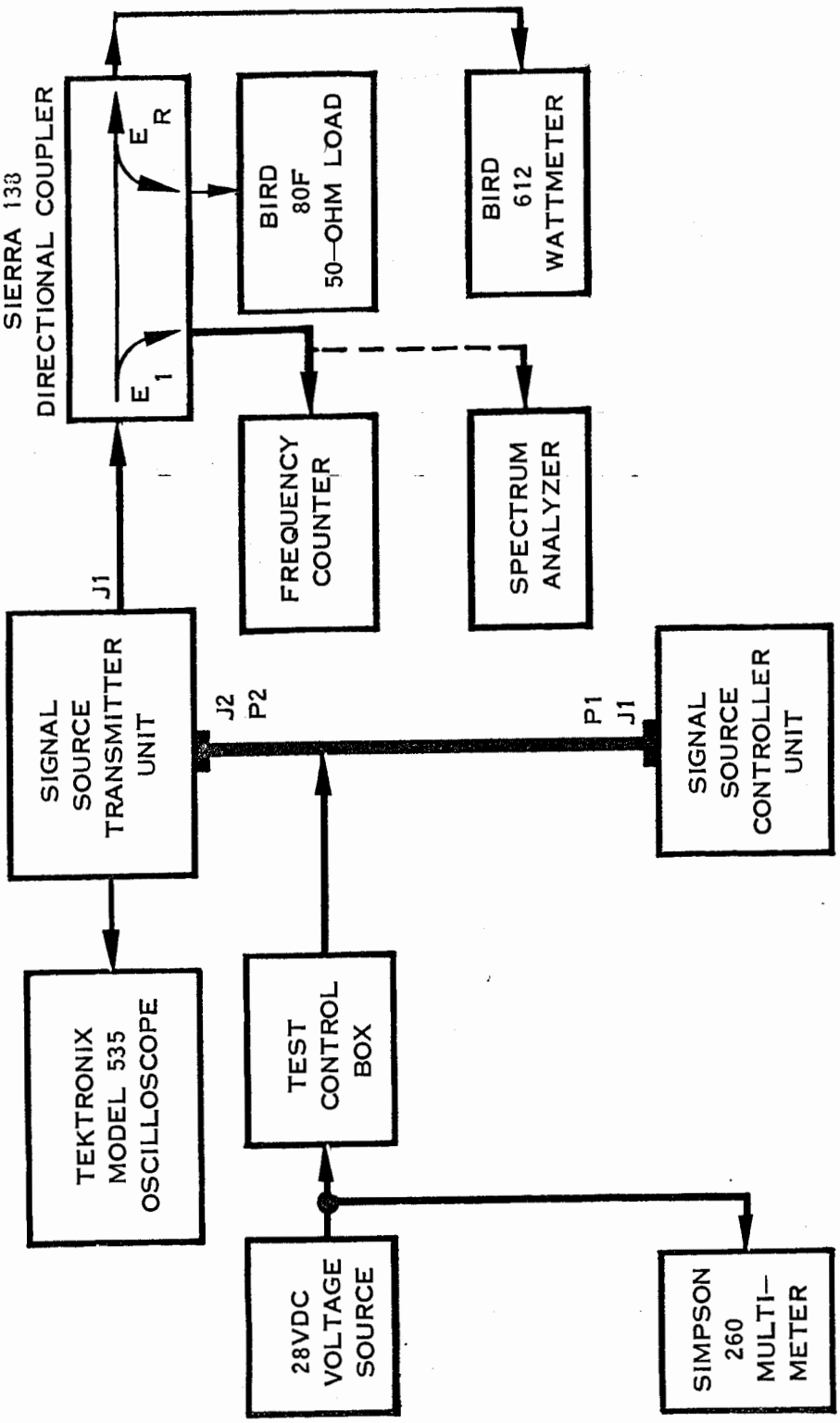


Figure 2. Test Equipment Setup For a Matts Signal Source

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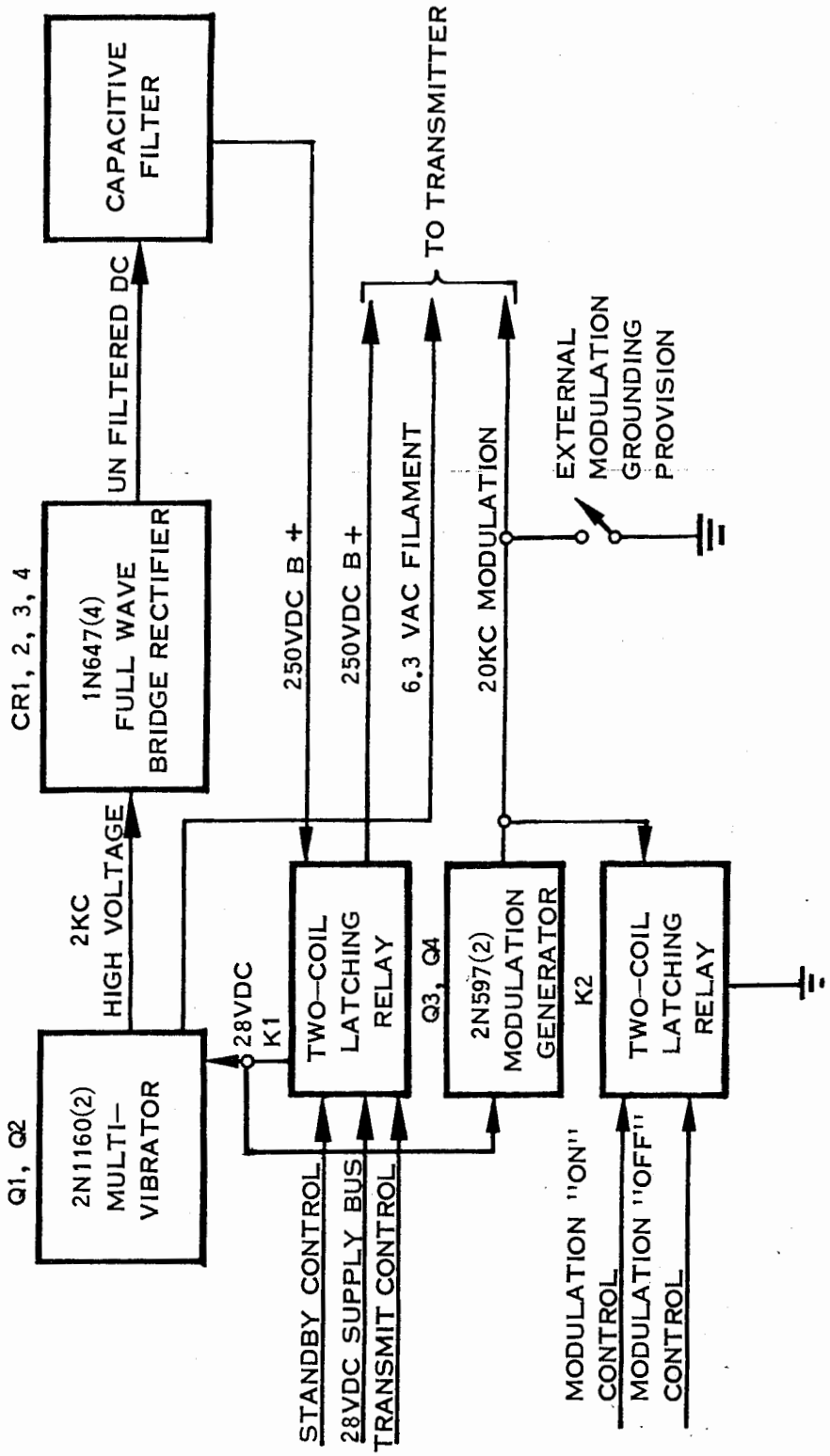


Figure 3. Block Diagram of the Controller Unit

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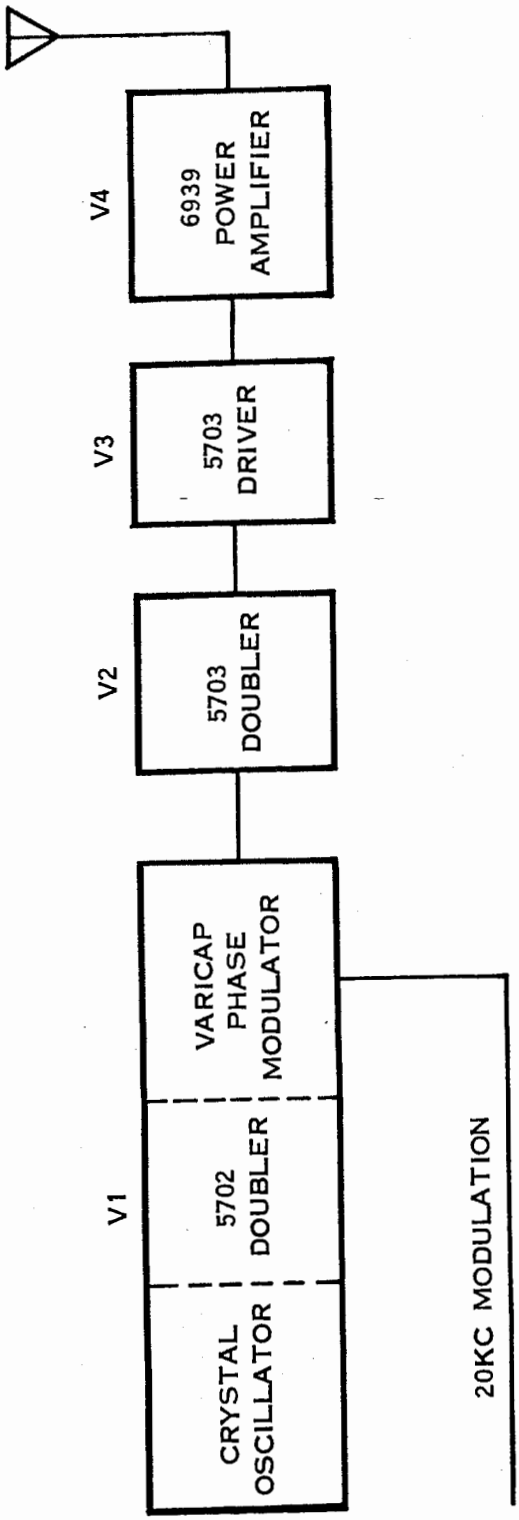


Figure 4. Block Diagram of the Transmitter Unit

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*4.5.2.5 Modulation index. The test shall be conducted concurrently with test 4.5.2.4 and at the dc input specified herein. Measure first order sideband amplitude and calculate modulation index (figure 5). Minimum index permitted shall be 0.7.

*4.5.3 Environmental tests

4.5.3.1 Immersion test. Completely immerse the transmitter group in previously boiled water at a temperature between 11° centigrade (C) and 20°C. A wetting agent, to reduce surface tension, may be dissolved in the water. Reduce the absolute pressure of the air above the liquid to 1 inch mercury (Hg) and maintain for 1 minute or until air bubbles substantially cease to be given off by the liquid, whichever is longer; then increase the air pressure to 2.5 inches (Hg) for a minimum period of 3 minutes. Any bubbles coming from within the transmitter or controller case shall be considered as leakage and cause for rejection. Bubbles that are the result of entrapped air on the various exterior parts of the transmitter group shall not be considered as leaks. Units shall be disassembled and inspected following this test to confirm absence of moisture. Units shall be rejected if any evidence of moisture leaks are found.

4.5.3.2 Vibration test. This test shall consist of sinusoidal vibration for a period of 54 minutes in each of three mutually perpendicular axes, with the frequency sweep applied at a constant logarithmic or linear rate, from 5 to 2000 Hz in 9 minutes, and back to 5 Hz in an additional 9 minutes. Also, it is required that the vibration be applied for 18 minutes in each of the three axes prior to completing the additional 36 minutes per axis. The control accelerometer shall be mounted as close as possible to the base of the test specimen. The vibration amplitudes shall be as follows:

Frequency	Amplitude
5-15.5 Hz	0.4 displacement amplitude (D.A)
15.5-200 Hz	5.0 gravities (g), vector
200-2000 Hz	7.0g, vector

The equipment shall operate in accordance with specifications before, during and after completion of vibration tests. The carrier signal will be continuously modulated during the entire test. The following parameters shall be monitored before, during and after test:

Parameter	Tolerance
28 volts direct current (vdc) input	± 1 volt
Carrier frequency	221 MHz ± 11 KHz (interceptor) 224 MHz ± 11 KHz (drone) 225 MHz ± 11 KHz (missile)
Modulated index	0.7 minimum
Modulation frequency	20 KHz ± 250 Hz

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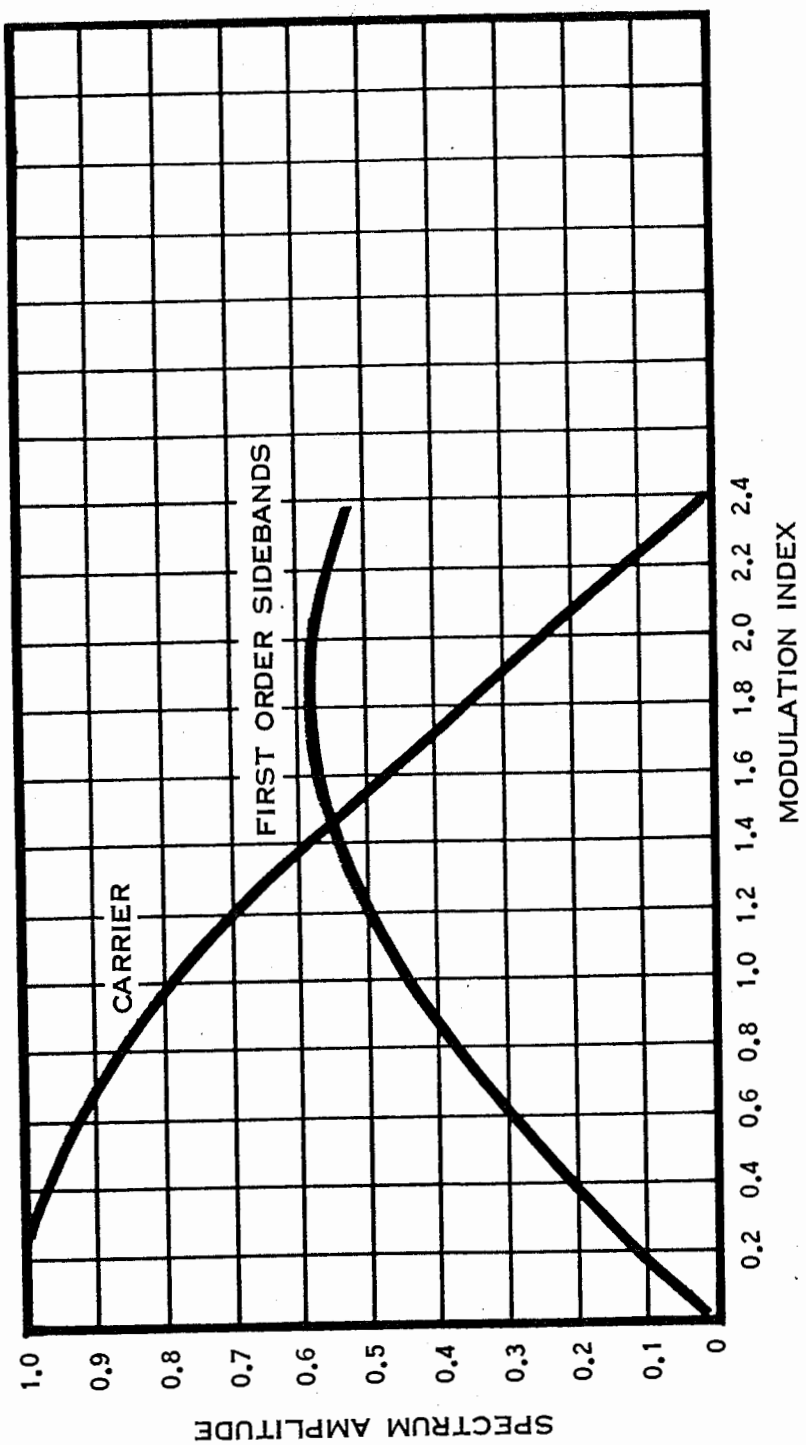


Figure 5. Graph For Determining the Modulation Index

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Parameter	Tolerance
Modulation voltage	3.2 volts alternating current (ac) minimum
Power output (modulated)	minimum 7 watts into a 50 ohm load

*4.5.3.3 Temperature test. This test consists of cooling the transmitter to +40°F and accomplishing the following bench checks without warm-up of the transmitter:

a. Turn the B+ and filament voltages on simultaneously. An RF power output is required 30 seconds after applying power. The rated power output of 7 watts is required one minute after applying power.
b. Turn power (B+ and filament) off for 5 seconds. Turn power (B+ and filament) on simultaneously. RF output power shall be 7 watts in 5 seconds.

*4.6 Inspection of preparation for delivery. Preservation, packaging and packing shall be inspected for conformance to the requirements of section 5.

*5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-E-17555.

5.1.1 Level A. Transmitter group shall be preserved and packaged in accordance with MIL-P-116, Method II B.

5.1.2 Level C. Transmitter group shall be preserved and packaged to a degree which will afford protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This level, in general, will conform to applicable carrier rules and regulations. Suppliers commercial practices may be used when they meet the requirements of this level.

5.2 Packing

5.2.1 Level A. Units preserved and packaged in accordance with 5.1.1 shall be packed in exterior type shipping containers conforming to PPP-B-601. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto.

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5.2.2 Level B. Units preserved and packaged in accordance with 5.1.1 shall be packed in exterior type shipping containers conforming to PPP-B-636. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Containers shall be sealed with tape conforming to PPP-T-60 in accordance with the appendix thereto.

5.2.3 Level C. Packages which require over-packing for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet consolidated freight classification rules and regulations of other common carriers as applicable to the mode of transportation.

6. NOTES

6.1 Intended use. The transmitter Group OA-2857/GSQ-29 is intended for use in interceptor aircraft, drone targets, and AIR-2A rockets in conjunction with Scoring System, Vector Miss Distance AN/GSQ-29.

*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Number of preproduction samples and point of testing.
c. Carrier frequency required for specific mode of operation as follows:
   (1) 225 mhz (AIR-2A) Crystal Number CR-54/U-1 (Cubic Part Number 19052-1).
   (2) 224 mhz (Drone) Crystal Number CR-54/U-2 (Cubic Part Number 19052-2).
   (3) 221 mhz (Interceptor) Crystal Number CR-54/U-3 (Cubic Part Number 19052-3).
d. The applicable level of packaging and packing required.

6.3 Special service tools. This section presents the uses, descriptions, and part number designations of the special equipment required to maintain the transmitter group.

6.3.1 Descriptions. The total special equipment for the transmitter group consists of two units. These are a modified casing, Cubic Part Number 24266, or equal; and a tuning tool, Cambridge Thermionic Corporation, Part Number CTC 2375, or equal. Figures 6 and 7 are figures of these items.

6.3.2 The casing has been modified from standard items to provide access openings for internal adjustments of the transmitter unit. The tuning tool is designed to minimize body capacitance effects of the technician during tuning operation.

6.3.3 Uses. The equipment described above is used during adjustment and alignment procedures of the transmitter unit. The alignment casing is installed in place of standard items and duplicates capacitive, inductive, and shielding effects of the normal configuration.

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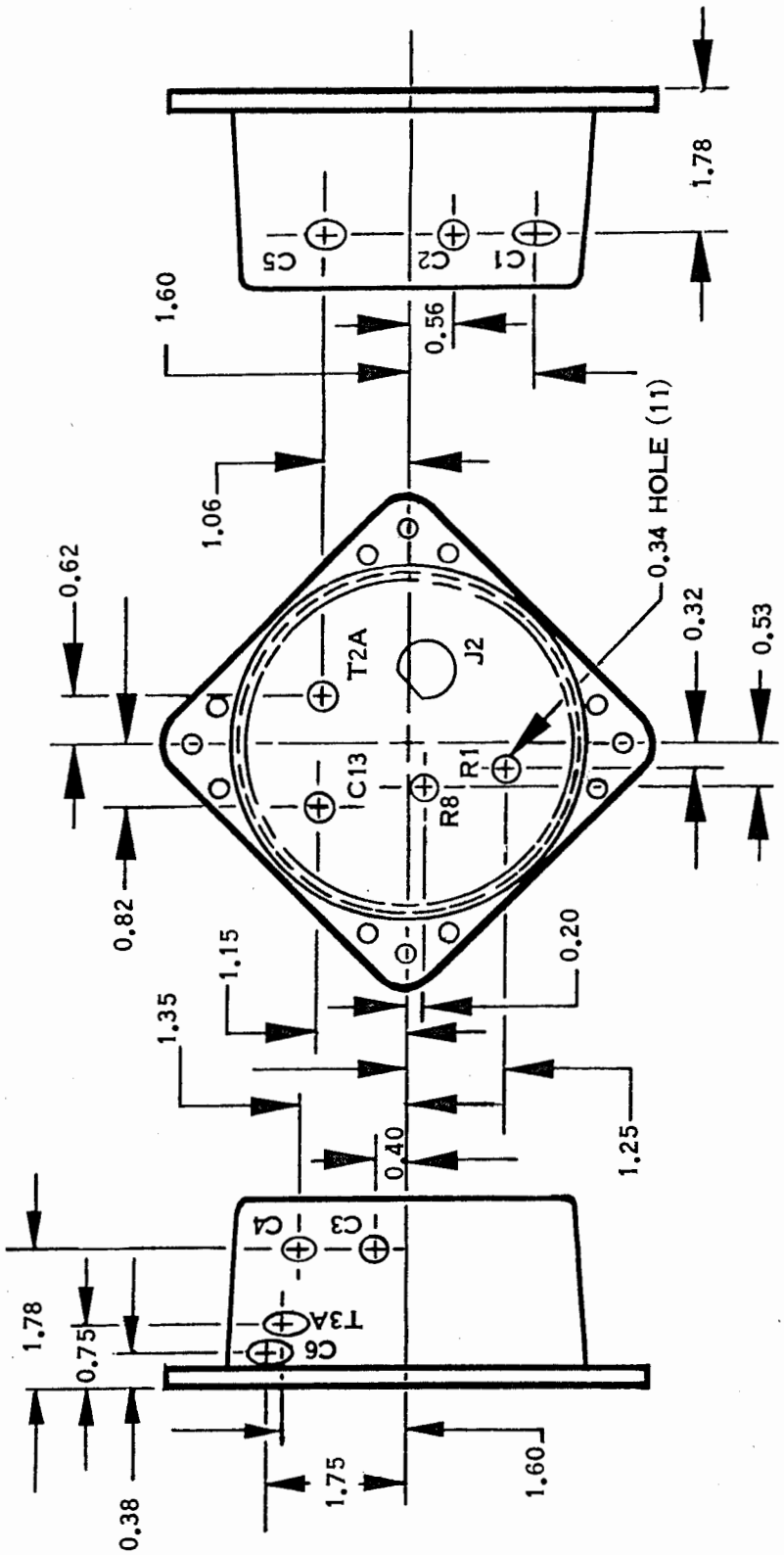
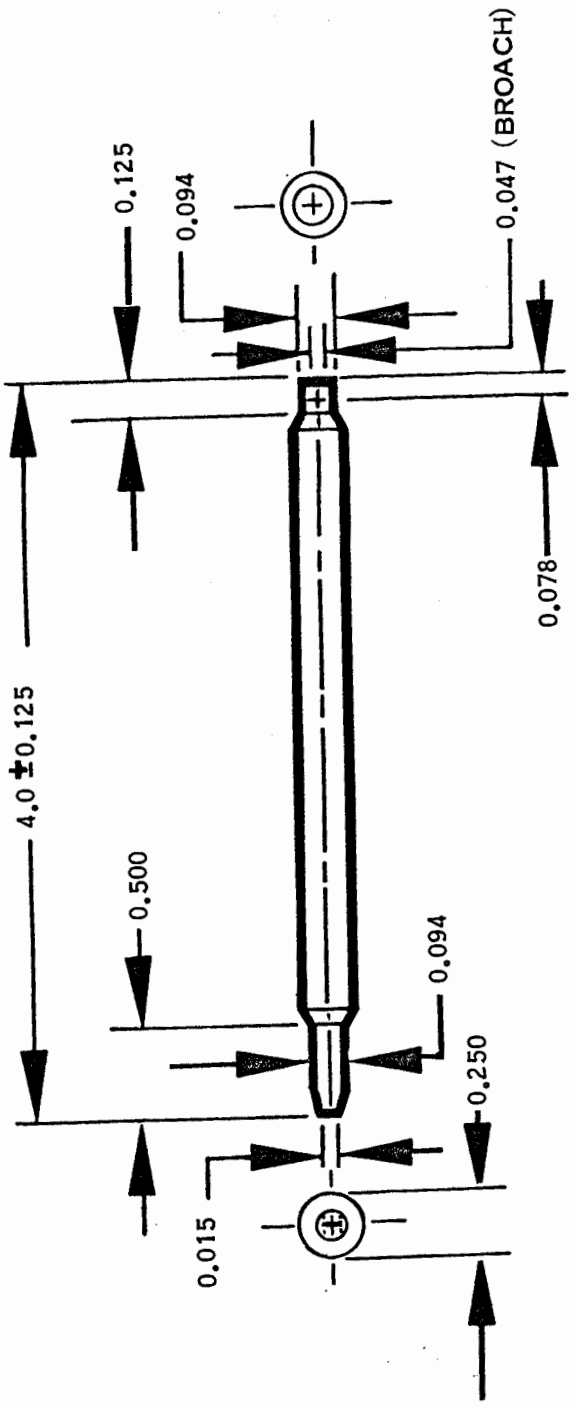


Figure 6. Alignment Casing

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*6.4 Asterisks. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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